Functions (Part 1)

PreCalculus - I

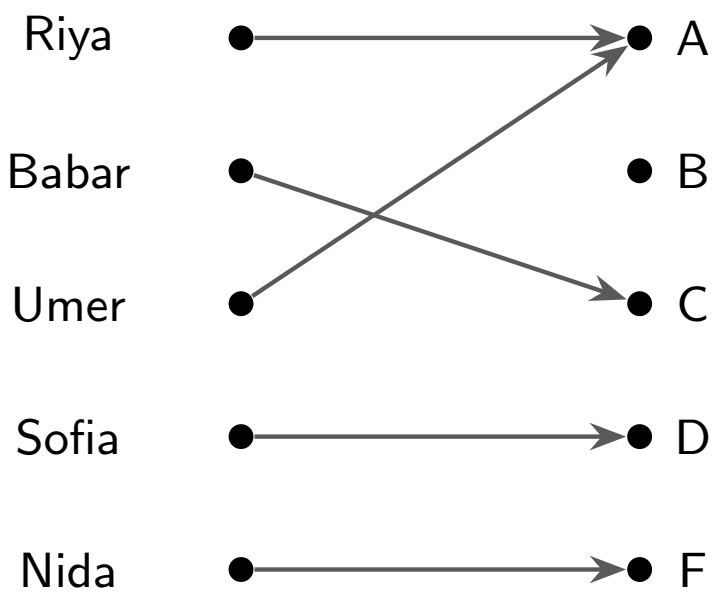
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26CS (Evening) - Fall 2026

What is a Function?

Intuition: In many situations we assign to each element of a set a particular element of a second set.

Example: Each student in a QR course is assigned a letter grade from $\{A, B, C, D, F\}$.



Formal Definition of a Function

Let A and B be nonempty sets.

Definition: A function f from A to B is an assignment of exactly one element of B to each element of A.

- If b is the unique element of B assigned to $a \in A$, we write $f(a) = b$.
- If f is a function from A to B, we write $f : A \to B$.

Key idea: A function takes an element from a set and maps it to a unique element in another set.

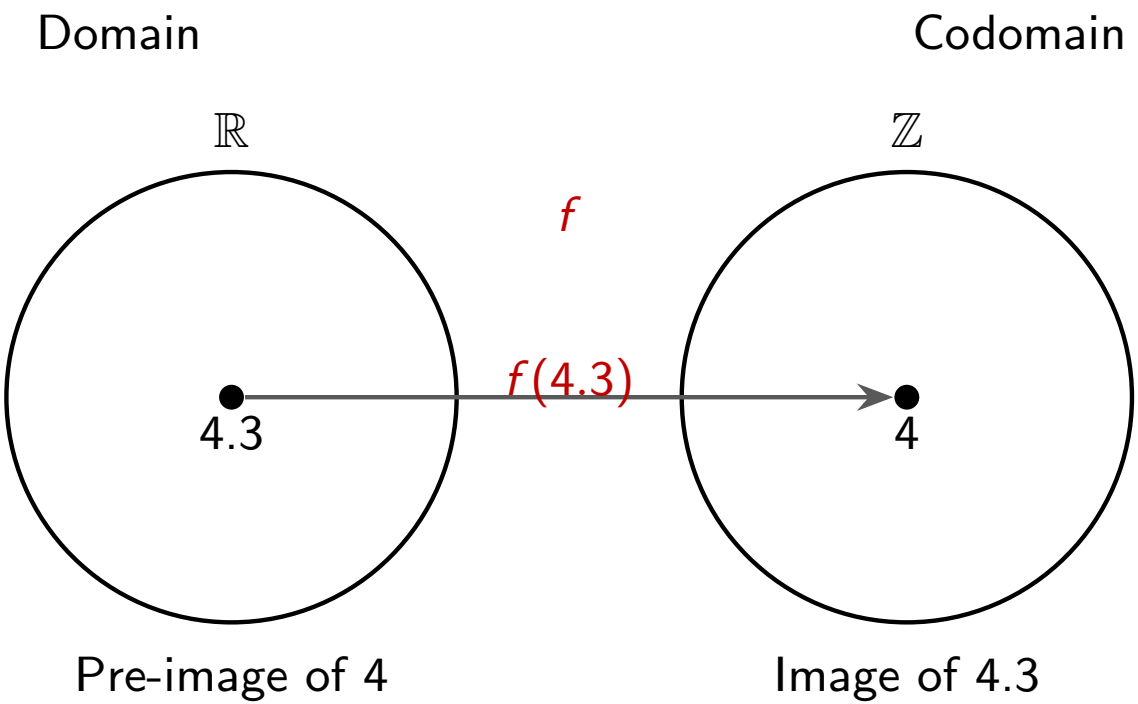
Domain, Codomain, Range

If $f : A \to B$:

- A is the domain of f.
- B is the codomain of f.
- If $f(a) = b$, we say b is the image of a, and a is the pre-image of b.
- The range of f is the set of all images of elements of A.
- We say f *maps* A to B.

Note: The range is always a subset of the codomain; it need not equal it.

Domain / Codomain Diagram



"f maps $\mathbb{R}$ to $\mathbb{Z}$" — but not every real number maps to itself.

Arrow Diagram of a Function

The definition of a function implies that the arrow diagram for a function f has two properties:

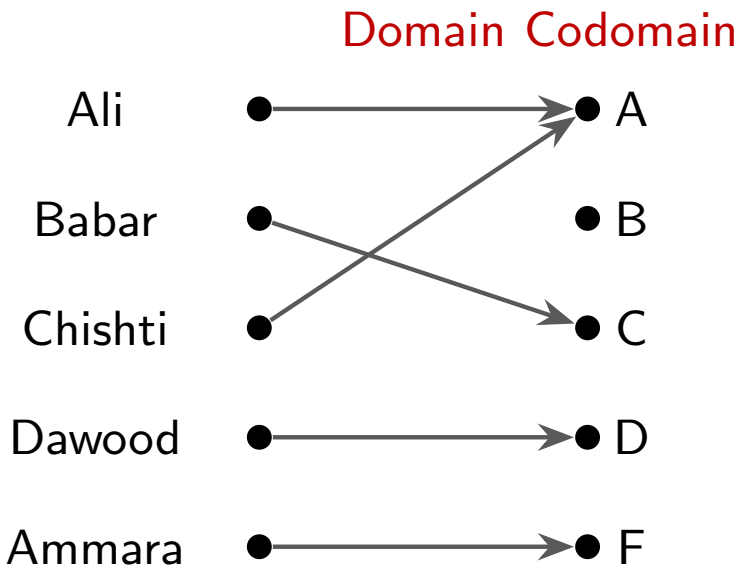
1. Every element of A has an arrow coming out of it.
2. No element of A has two arrows coming out of it that point to two different elements of B.

Note

Failure of either property means the diagram does not represent a function.

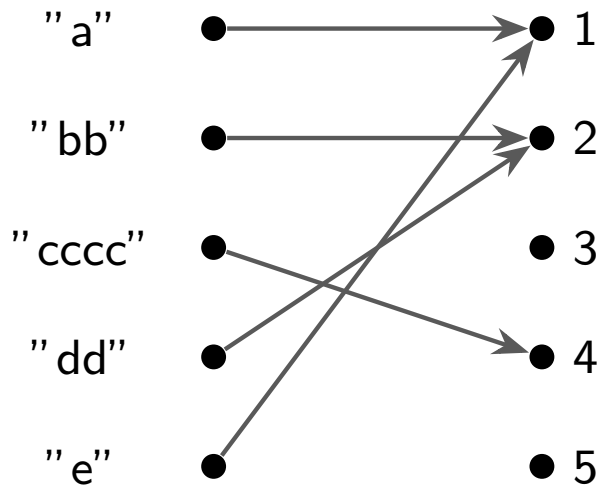
Arrow Diagram Examples

Class grade function



$g(\text{Ali}) = A,\ g(\text{Babar}) = C,\ g(\text{Chishti}) = A, \ldots$

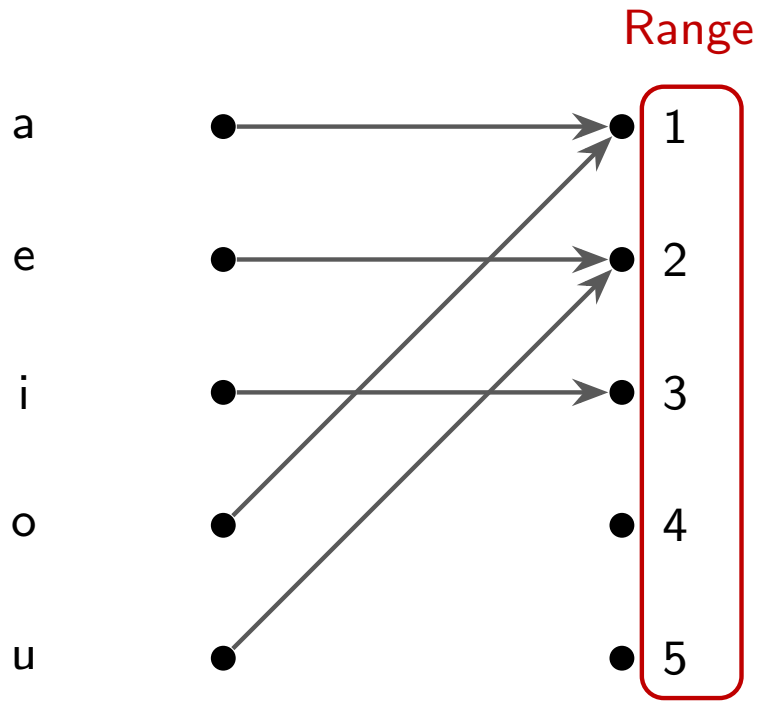
String length function



$f(x) =$ length of x, e.g.
$f(\text{"a"}) = 1,\ f(\text{"bb"}) = 2.$

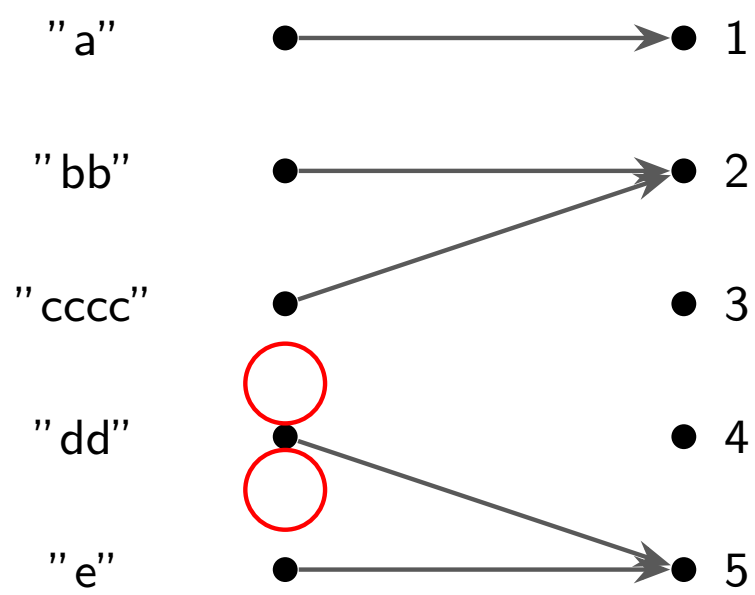
The Range of a Function

The range of f is the set of all images of elements of A.



Range $= \{1, 2, 3\}$, while codomain may be larger.

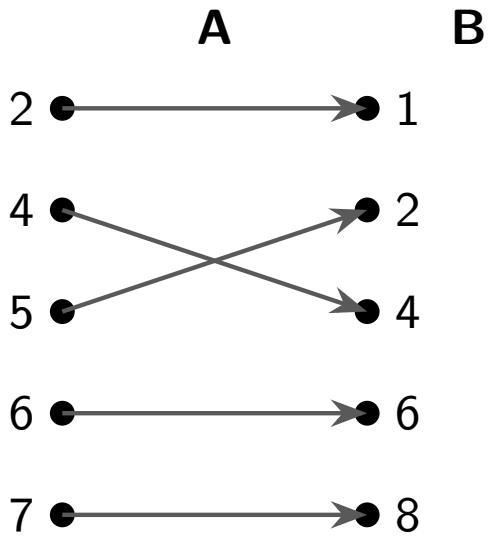
Not a Valid Function



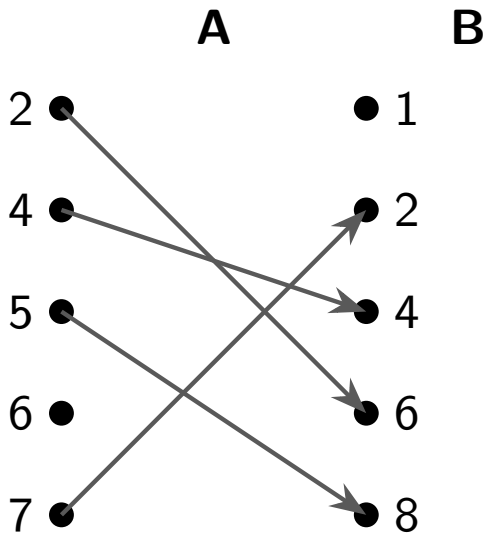
Not a valid function: the elements "dd" and "e" both map to two different values.

Function vs. Non-Function

Which of the following arrow diagrams define functions from $A = \{2, 4, 5, 6, 7\}$ to $B = \{1, 2, 4, 6, 8\}$?



Not a Function



Not a Function

Note

Every element of the domain must map to exactly one element of the

Example: Squaring Function

Let $f : \mathbb{Z} \to \mathbb{Z}$ assign the square of an integer to this integer.

- $f(x) = ?$ $\boxed{f(x) = x^2}$
- Domain: the set of all integers ($\mathbb{Z}$).
- Codomain: the set of all integers ($\mathbb{Z}$).
- Range: $\{0, 1, 4, 9, 16, \ldots\}$ — all perfect squares.